

Functional Groups

Answer key with worked solutions

A. Understanding

1. Functional groups.

(a) An **atom or group of atoms** that gives a family of organic compounds its typical reactions and properties. (b) Both contain the **same functional group (-OH)**; reactions happen at the functional group, and the different chain length has little effect.

B. Identifying and naming

2. Functional groups.

(a) **alcohol** (b) **aldehyde** (c) **ketone** (d) **carboxylic acid** (e) **ester** (f) **amine** (g) **haloalkane** (chloro) (h) **amide**

3. Names.

(a) **propan-1-ol** (b) **propanal** (c) **butanone** (butan-2-one) (d) **methanoic acid** (e) **propanoic acid** (f) **2-chloropropane**

4. Condensed formulas.

(a) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$ (c) $\text{CH}_3\text{CH}_2\text{COOCH}_3$ (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

C. Isomers and properties

5. Functional group isomers.

(a) **propanal**, $\text{CH}_3\text{CH}_2\text{CHO}$ (aldehyde), and **propanone**, CH_3COCH_3 (ketone). (b) **ethanol**, $\text{CH}_3\text{CH}_2\text{OH}$ (alcohol), and **methoxymethane**, CH_3OCH_3 (ether).

6. Boiling points.

Propane has only weak dispersion forces. **Methoxymethane** is polar (C–O bonds), so it has dipole–dipole forces too, but no O–H bond, so no hydrogen bonding. **Ethanol** has an O–H group and forms **hydrogen bonds** between molecules, so much more energy is needed to separate them. **Ethanoic acid** has an O–H and a C=O group, forms **stronger and more hydrogen bonds** (molecules pair up), and is also heavier, so it boils highest.

D. Esters

7. Esters.

(a) **methyl ethanoate**: $\text{CH}_3\text{COOH} + \text{CH}_3\text{OH} \rightleftharpoons \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$ (b) **ethyl propanoate**, $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$ (c) **methanoic acid** (HCOOH) and **propan-1-ol** ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$)

8. Yield of methyl ethanoate.

(a) $n(\text{CH}_3\text{OH}) = \frac{4.00 \text{ g}}{32.04 \text{ g/mol}} = 0.1248 \text{ mol}$; the ratio is 1 : 1, so $n(\text{ester}) = 0.1248 \text{ mol}$
 $m = 0.1248 \text{ mol} \times 74.08 \text{ g/mol} = \mathbf{9.25 \text{ g}}$

(b) Percent yield = $\frac{6.50 \text{ g}}{9.25 \text{ g}} \times 100 \% = \mathbf{70.3 \%}$. The reaction is **reversible** and reaches equilibrium, so not all the methanol is converted (some product is also lost when it is separated and purified).

E. Challenge

9. Identifying the liquid.

(a) In 100 g: $n(\text{C}) = \frac{54.53 \text{ g}}{12.01 \text{ g/mol}} = 4.540 \text{ mol}$; $n(\text{H}) = \frac{9.15 \text{ g}}{1.008 \text{ g/mol}} = 9.077 \text{ mol}$; $n(\text{O}) = \frac{36.32 \text{ g}}{16.00 \text{ g/mol}} = 2.270 \text{ mol}$

Divide by the smallest (2.270 mol): C 2.000, H 3.999, O 1, so the empirical formula is $\text{C}_2\text{H}_4\text{O}$ (44.05 g/mol).

$\frac{88.1 \text{ g/mol}}{44.05 \text{ g/mol}} = 2.00$, so the molecular formula is $\text{C}_4\text{H}_8\text{O}_2$.

(b) $\text{C}_4\text{H}_8\text{O}_2$ with a sweet smell and no reaction with carbonate: an **ester**, for example **ethyl ethanoate**, $\text{CH}_3\text{COOCH}_2\text{CH}_3$ (methyl propanoate and propyl methanoate are also correct).